Parallel Physics-Informed Prognostics for EV Battery Health Management: Fusing Physics Constraints, Uncertainty and GPU-Scale Inference

A. Vibhas Chowdhary · Maddala Jithendra — *guide: Dr. Kavitha R*

Department of Computing Technologies, SRM Institute of Science and Technology, Kattankulathur · va1641@srmist.edu.in · jm4408@srmist.edu.in · kavithar14@srmist.edu.in

BACKGROUND & RESEARCH GAP 01

Battery RUL research has four mature ingredients — but **no published framework combines them**. Each thread solves one piece and stops there; a maintenance decision needs all four at once.

approach family	physics-informed	RUL distribution	GPU-scale	real-cell benchmark
Physics-informed NNs [13], [22]	✓	✗	✗	partial
LSTM + particle filter [27], [28]	✗	✓	✗	single cell
CUDA-parallel particle filters [17]	✗	✓	✓	not battery
P3-EVHM — this work	✓	✓	✓	✓

The gap we fill: verified against a systematic review of 33 works — the closest LSTM+filter hybrid [28] explicitly lists our physics constraints as *future work*; GPU particle filters [17] never touched battery data. **P3-EVHM closes all four cells at once, on real NASA PCoE cells, across two machines.**

THIS WORK — THE BREAKTHROUGH IN THREE CONTRIBUTIONS 02

C1 — Physics-informed training that removes impossible predictions. Two electrochemistry laws enter the loss (Eq. 4–5): capacity may never rise, and decline must respect an Arrhenius-style temperature prior. Implausible (rising-capacity) steps fall **0.366 → 0.076** — a ~5× cut, with better capacity MAE (0.285 → 0.257 Ah).

C2 — RUL as a calibrated distribution, not a guess. The trained model seeds thousands of simulated futures (Eq. 6–8) reweighted by the measured capacity; we report the median, the 95% interval, and decision rules driven by its lower bound: **a RUL estimate in 20/20 configurations on both machines** (mean error 18.8 cycles, interval coverage 0.85).

C3 — GPU-scale particle inference, open and reproducible. The cloud updates as one vectorised tensor op per step (SIMT): **16.5× end-to-end at 500,000 particles**, sub-100 ms device time throughout — shipped as an MIT framework (pip, dashboard, CI, Kaggle kernel) with 40 measured configurations.

METHOD — THE PHYSICS ENTERS THE LOSS 03

(4) $L = MSE(\hat{q}, q) + \lambda_1 \cdot mean\ ReLU(\hat{q}_t - q_t) + \lambda_2 \cdot mean(\Delta\hat{q}_t - g(T_t))^2$

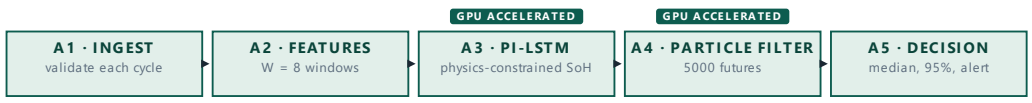
(5) $g(T) = clamp(\hat{A} \cdot (T - \bar{T}), min = 0)$ — *prior fitted from the training cycles, not a hard clamp*

(6) $q_i^{(k)} = q_i^{(k-1)} - d_i + \varepsilon_k, \quad w_i \propto exp(-(q_{obs} - q_i)^2 / 2\sigma^2)$

(8) $RUL = median\{k : q_i^{(k)} \le q_{EOL}\}, \quad report\ [P_{2.5}, P_{97.5}]$

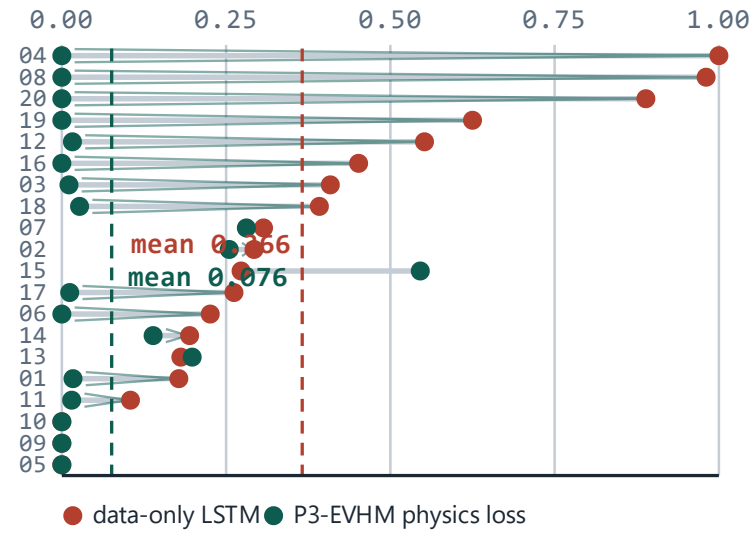
A 1–32 LSTM over a W = 8 window; identical seeds and metrics for every baseline B1–B6 so the comparison is the physics, not the tuning.

SYSTEM — FIVE AGENTS, ONE FRAMEWORK 04



A3 and A4 run on the GPU; the same code runs on CPU for parity. Four doors into the same engine: Python API, CLI, dashboard, Kaggle kernel.

EVIDENCE 1 — THE PHYSICS LOSS WORKS 05



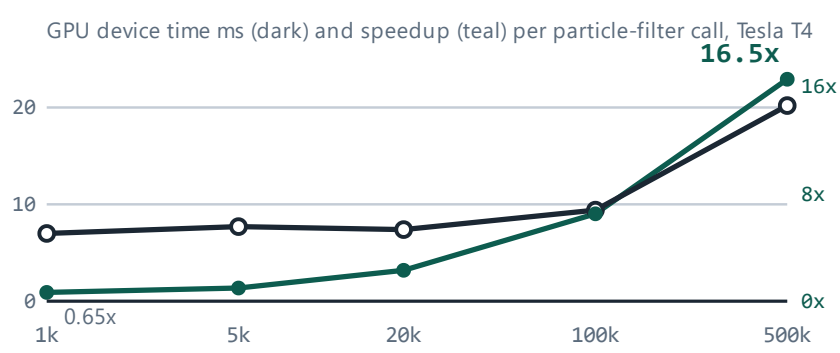
Violation rate V per configuration (share of predicted steps where capacity *rises* — physically impossible). Every one of 20 configurations improves or ties; the two catastrophes (V = 1.0, configs 04/08) are eliminated entirely.

EVIDENCE 2 — PROGNOSTIC QUALITY 06

QUANTITY	MEASURED	NOTE
RUL produced	**20 / 20**	every config, CPU *and* T4
mean RUL error	18.8 cyc	18.6 on CPU — reproducible
95% interval coverage	0.85	honest, below the 0.95 ideal
GPU speedup @500k	16.5×	20.2 ms per PF call

Transfer (leave-one-cell-out): a 21 KB pretrained checkpoint, body frozen, head fine-tuned for 0.2 s — RUL in **4/4 held-out cells** vs **1/4** from scratch, zero physics violations vs 0.355 max.

EVIDENCE 3 — GPU SCALING 07



GPU device time ms (dark) and speedup (teal) per particle-filter call, Tesla T4

Honest crossover reported: below ~5k particles the warm CPU wins (0.65×); the GPU pays exactly where prognostics needs it — large clouds, sharp intervals, real time.

CONCLUSIONS & FUTURE WORK 08

Conclusions. The fusion is the contribution: physics constraints make LSTM prognostics physically consistent at no accuracy cost; the particle filter turns that forecast into a calibrated maintenance decision in every configuration tested; and GPU-scale inference makes the cloud size that sharp intervals need affordable. Open-sourced end to end, with 40 measured configurations as evidence.

Future work. Pack-level validation beyond single cells; prior-floor tuning for early-start predictions (40% of life); interval coverage toward 0.95; streaming on-vehicle deployment.

Code & full validation: github.com/Jithendra123892/p3-evhm (MIT) · GPU kernel: jithendramaddala/p3-evhm-gpu-run **Reproduce:** pip install -e . · python -m p3evhm.demo · streamlit run app.py

Key references [1] Saha & Goebel, *Battery Data Set*, NASA PCoE, 2007. [2] Arulampalam et al., *IEEE Trans. Signal Process.* 50(2), 2002. [13] Shi, Rivera & Wu, *Mech. Syst. Signal Process.* 179, 2022. [17] *Parallel accelerated CRPF on CUDA*, Appl. Intell., 2020. [22] do Nascimento et al., *Sci. Rep.* 13, 2023. [27] *AI-driven particle filter for battery RUL*, Batteries 10(12), 2024. [28] *Hybrid particle-filter-trained neural network*, IEEE, 2021. — full 33-entry bibliography in the paper; every figure generated from `results/kernel_grid_stats.json`.